

Cansat Competition Guide 2027

PocketQube

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Revisions

Date	Changes
11/20/2025	Initial guide
2/10/2026	Cleaned up document deliverable description. Removed dates from various parts of the document and added references to the calendar in the appendix. Cleaned up some requirements.
3/4/2026	Added XBEE radio as an optional back up radio link.
3/18/2026	Updated design requirements to have two canisters per rocket. Teams do not design nose cones. Updated container design in appendix. Numerous requirements have been updated or deleted for this new configuration.
4/28/2026	Updated requirements for the container. Added ring to top side and changed parachute attachment method.
6/8/2026	Update to require a power switch or RBF for container power to be activated by judge only. Beacon to be a power indicator, not software controlled. Pocketqube to incorporate a beacon and turn on when pocketqube is powered. Not software controlled.
6/16/2026	Added thermal requirement to M10. Updated CTR29. Moved sections around.
7/22/2026	Updated requirement CTR28.
8/3/2026	Updated C11 to clarify the axis of rotation for the camera view.
8/23/2026	Updatred Appendix F with clarifications about the wood disks. Corrected dates in table 1.
9/3/2026	Added payment start date in schedule appendix.
9/15/2026	Corrected PDR and CDR presentation times.
9/30/2026	Updated requirement PQ7 adding a minimum width for the solar panels.

Introduction

The Cansat competition is a design-build-fly competition that provides teams with an opportunity to experience the design life-cycle of an aerospace system. The Cansat competition is designed to reflect a typical aerospace program on a small scale and includes all aspects of an aerospace program from the preliminary design review to post flight review. The mission and its requirements are designed to reflect various aspects of real world missions including telemetry, communications, and autonomous operations. Each team is scored throughout the competition on real-world deliverables such as schedules, design review presentations, and demonstration flights.

Mission Overview

The mission simulates the design of a satellite and deployment system. The specifications are based on the PocketQube specifications. One deviation from the PocketQube specification is the need for a parachute to be attached to the PocketQube. This mission focuses on mechanisms, systems engineering, and satellite operations. The challenge is to develop a compact satellite with multiple mechanisms and multiple sensors.

The CanSat shall consist of a container and a PocketQube satellite (Requirements CTR1, PQ1, & PQ2). Two containers shall be flown on one rocket (Requirement CTR3). This simulates a ride share.

Teams will design a Cansat which includes a container that slides into the rocket payload airframe and contains a PocketQube. The container shall secure the PocketQube using a rail and mechanism to hold the PocketQube in place until it is to be released (Requirements CTR6, CTR10, CTR11, CTR12). The container shall transmit telemetry at 4 Hz when turned on (Requirement CTR15). After launch, when the rocket deploys its parachute at peak altitude, the containers are ejected from the rocket payload section and shall descend on a parachute at 10 m/s (Requirements C1, C2, & CTR5). When the container descends to 90% of peak altitude, it shall release the PocketQube (Requirement C3). Five seconds after the PocketQube is released, the container shall reduce its telemetry rate to 1 Hz and stop transmitting when it lands (Requirement CTR16).

While the PocketQube is in the container, it shall not be powered (Requirements CTR7 & PQ3). When released from the container, the PocketQube shall power on and start transmitting telemetry at 4Hz (Requirements C4, C5, PQ21, & PQ22). Five seconds after release, the PocketQube shall deploy two solar panels (Requirements C6, PQ7, PQ8). The solar panels shall generate a voltage to be measured (Requirement PQ26). It is not required to power any electronics. A recording video camera shall capture the ground pointing in the

nadir direction (Requirements C7, PQ13, & PQ14). The PocketQube shall maintain the video orientation so the video of the ground is not rotating (Requirement C8 & PQ15). The PocketQube descent rate shall be 3 m/s to 5 m/s (Requirement C13).

The PocketQube shall listen for commands from the ground station (Requirement C7). A command shall be transmitted to deploy a boom that is twice as long as the length of the PocketQube with a magnetometer at the end (Requirements C9, PQ9, PQ10, & PQ11). The deployed boom shall be positioned so it is in the recording camera view (Requirements C10 & PQ12). After 30 seconds of flight, a command shall be transmitted to rotate the camera view 90 degree along the nadir axis in the direction specified by the ground station judge (Requirement C11, PQ16, PQ17, & PQ18).

Teams may include a science experiment in their PocketQube of their own design as a bonus objective (Requirements B1, B2, B3, B4, & B5).

https://www.albaorbital.com/pocketqube-standard

Team Selection

Since the competition is sponsored by US organizations, the competition is required to include at least 20 US teams. During the application phase of the competition, up to 20 teams per country can participate. If more than 20 apply per country, a random selection will be made for the 20 teams per country.

From the top 50 selection, 20 teams are required to be from the US. The remaining teams can be from other countries. Teams from each country will be ranked only among their country. Then the top scoring teams from each country will be selected in a round robin method. The top team from each country will be selected in the first round, then the next top team from each country will be selected in the second round and the selection is repeated until 50 teams are selected. This will create the greatest diversity in teams and allow more countries to participate.

Competition Description

To control the size of the competition, only three teams per school are allowed to apply to the competition. It is recommended that schools hold internal design competitions to determine the three teams to apply. If more than three teams from one school apply, the first three applications received will be accepted.

The competition is in five phases.

Phase One

Phase one is the application phase. Teams must submit an application and a $300 competition fee that is non-refundable. The fee is used to offset the cost of rocket motors and other materials. Applications must be submitted by the due date in the calendar appendix. Payments will be requested in early November and must be paid by the given due date in the calendar.

Phase Two

Phase two is the preliminary design. Teams are to develop designs, prototype, test concepts and generate a preliminary design review (PDR) slide package using the provided template. Teams will submit PDR slides by the designated due date. Slides will only be accepted in PDF format. Teams that do not meet the due date or do not submit in the proper PDF format will be dropped from the competition. Teams will be given a calendar of available time slots to select for presenting their PDR. Teams will have 40 minutes to discuss a subset of the PDR slides via telecon.

After the PDR reviews have been completed, the score sheets will be reviewed for completeness and potential errors before being made available to teams. Teams will have a short time period to review and identify potential errors. Teams must submit errors within the review period. The identified errors will be reviewed and if agreed, the score sheet will be updated. Afterwards, a total of 50 teams will be invited to the competition.

Phase Three

Phase three is the critical design. Teams will finalize their design and start ordering components, manufacturing parts, test subsystems and start developing the flight unit. Teams will generate a critical design review (CDR) slide package using the provided template. Teams will submit CDR slides by the designated due date. Slides will only be accepted in PDF format. Teams that do not meet the due date or do not submit in the proper PDF format will be dropped from the competition. Teams will be given a calendar of available time slots to select for presenting their CDR. Teams will have a half hour to discuss a subset of the CDR slides via telecon.

Phase Four

Phase four is the launch weekend. Friday, teams will be scheduled to present their completed Cansat for flight readiness review which must be completed in 30 minutes. The Cansat must be ready to launch at this time. It must be completely assembled and flight ready. Each team will be scored during the flight readiness review. Teams can only present once at the flight readiness review at their designated time. Teams late for the review will lose points. CanSats must pass the drop test, fit check, and battery verification in order to fly. Multiple attempts at the drop test are allowed until the flight readiness reviews are completed.

Saturday is launch day; teams will perform final preparations and turn in their Cansats by 12:00 hours local time. Launch will start at 13:00 hours local time and continue until all launches are completed. There will be no second flights unless the fault is of the launch

provider and there are spare rockets and rocket motors.

Phase Five

Phase five is the Post Flight Review (PFR). Post Flight Review is a 15 minute presentation of the flight results and 5 minutes for questions. Awards will be presented at the end of all the post Flight Reviews.

For teams to receive certificates of accomplishment and be considered for awards, they must complete all phases of the competition.

Late submissions will not be accepted for phases one through three and the team will be dropped from the competition. Internet issues, file issues, email issues, and any other issues will not be considered. **Submit the documents early to be safe.**

All scoring and judging results are final. Scoring is set up to be quantitative with little qualitative scoring to minimize any biases.

The competition is operated by a dedicated group of volunteers who spend their own time supporting various phases of the competition. Some volunteers spend their own funds to attend and support the competition while others are graciously supported by their employers. The competition is designed to provide teams a great educational experience and to minimize the time of the volunteers. Strict due dates, file templates, and file name formats are required. Please follow all due dates and all submission requirements.

Requirements

Operational Requirements

Requirement Number	Requirement	Evaluation
C1	The Cansat shall be deployed from the rocket when the rocket motor ejection charge fires by sliding out of the airframe.	Observed during launch
C2	After deployment, the Cansat shall descend at 10 meters/second using a parachute that automatically deploys. Error is +/- 3 m/s.	Reported by the team during PFR using onboard sensors
C3	At 90% flight peak altitude, the PocketQube shall be released from the container.	Release observed during launch; release altitude reported by the team during PFR using onboard sensors

C4	The PocketQube shall power up after being released from the container.	Reported by the team during PFR using onboard sensors and/video
C5	The PocketQube shall start transmitting telemetry at 4 Hz.	Observed on ground station during launch
C6	Five seconds after the PocketQube powers up, it shall deploy two solar panels.	Observed on ground station during flight and reported by the team during PFR using onboard sensors
C7	The PocketQube shall support commands to turn on the video recorder, deploy the magnetometer boom, and rotate the camera view.	Observed on ground station during flight and reported by the team during PFR using onboard sensors
C8	The PocketQube shall record video in the nadir direction with less than +/- 10 degrees rotation in the video.	Observed on ground station during flight and reported by the team during PFR using onboard recording
C9	The PocketQube shall deploy a boom that is twice the length of the pocketcube on command.	Observed on ground station during flight and reported by the team during PFR using onboard sensors
C10	The boom shall be in view of the PocketQube camera when deployed.	Observed on ground station during flight and reported by the team during PFR using onboard recording
C11	The PocketQube shall rotate the video along the nadir axis at the specified angle on command.	Observed on ground station during flight and reported by the team during PFR using onboard recording
C12	The cost of the Cansat shall be under $1000. Ground support and analysis tools are not included in the cost of the Cansat. CanSat material from previous years shall be included in this cost, based on current market value.	Reported by teams during CDR
C13	The PocketQube descent rate shall be 3 m/s to 5 m/s.	Observed on ground station during flight and reported by the team during PFR using onboard recording

Container Requirements

Requirement Number	Requirement	Evaluation
CTR1	The container shall meet the mechanical dimensions and requirements specified in Appendix F.	Measured during FRR
CTR2	The container structure shall not be able to be deformed in a twisting manner or be compressible.	Assessed during FRR
CTR3	The container shall be designed to be installed in the rocket in any order. Teams cannot specify the stacking position of their CanSat.	Assessed during FRR
CTR4	The container shall include two vent holes no greater than 12 mm in diameter to support air pressure equalization.	Assessed during FRR
CTR5	The container shall include a parachute to control the descent rate to 10 m/s +/- 3 m/s that is attached and hangs loose in the rocket airframe.	Reported by the team during PFR using onboard sensors
CTR6	The container shall include the PocketQube deployment rail to hold the PocketQube in place.	Assessed during FRR
CTR7	The deployment rail shall depress the PocketQube power micro/limit switches when the PocketQube is stowed in the container so the PocketQube is not powered.	Assessed during FRR
CTR8	The walls surrounding the PocketQube shall be solid except for the vent holes.	Assessed during FRR
CTR9	The exit side of the container shall be fully covered so nothing can enter and tangle inside the container.	Assessed during FRR
CTR10	The container shall include a release mechanism for the PocketQube.	Assessed during FRR
CTR11	The container shall include a sensor to detect the release mechanism has fully	Assessed during FRR

	deployed.	
CTR12	The container shall include a sensor to detect that the PocketQube cleared the deployment rail.	Assessed during FRR
CTR13	The container shall record a video of the PocketQube release.	Assessed during PFR
CTR14	The container shall transmit telemetry specified in the telemetry section.	Assessed during FRR and reported by the team during PFR using onboard sensors
CTR15	The container shall transmit telemetry at a 4Hz rate when turned on.	Assessed during FRR and reported by the team during PFR using onboard sensors
CTR16	The container shall reduce the telemetry rate to 1 Hz five seconds after releasing the PocketQube.	Assessed during FRR and reported by the team during PFR using onboard sensors
CTR17	The container shall accept commands to trigger all deployment mechanisms.	Assessed during FRR
CTR18	The container shall monitor air pressure in Pascals to 1 Pascal resolution.	Reported by the team during PFR using onboard sensors
CTR19	The container shall determine its altitude in meters to 0.1 meter resolution using the air pressure sensor.	Reported by the team during PFR using onboard sensors
CTR20	The container shall monitor its internal temperature in C to 0.1C resolution.	Reported by the team during PFR using onboard sensors
CTR21	The container shall monitor its battery voltage to 0.01 volt resolution.	Reported by the team during PFR using onboard sensors
CTR22	The container shall monitor its battery current to 1 mA resolution.	Reported by the team during PFR using onboard sensors
CTR23	The container shall include a power switch or Remove Before Flight (RBF) power on switch that is easily accessible without the use of any tools to be turned on by the competition judge. Teams do not turn on their container.	Assessed during FRR and during pre-flight launch integration
CTR24	The power switch or RBF switch shall	Assessed during FRR

	be vibration and shock proof.	
CTR25	Markings shall be on the container to identify the location of the power switch or RBF switch and indicate the position for power on.	Assessed during FRR
CTR26	The container shall include an audio beacon that turns on when the container is powered on by the judge. It shall be the power on indicator.	Assessed during FRR and during pre-flight launch integration
CTR27	The PocketQube deployment end of the container shall not have any protrusions or features that can snag a parachute if a second container is on top.	Assessed during FRR
CTR28	The container shall accept the minimum required commands described in the telemetry and command section.	Assessed during FRR and at launch
CTR29	Both ends of the container shall be a 6 mm or ¼” thick plywood disk of 131 mm diameter.	Assessed during FRR
CTR30	The container parachute shall attach to the plywood disk using kevlar cord secured through two 5 mm holes that are positioned opposite of each other by 20 mm about the center of the disk.	Assessed during FRR
CTR31	When the CanSat is delivered for flight at the field, the CanSat shall be stored on a table sitting on the flat side with the parachute below.	Assessed during CanSat turn-in
CTR32	The team number shall be visible on the top side of the CanSat container.	Assessed during CanSat turn-in

PocketQube Requirements

Requirement Number	Requirement	Evaluation
PQ1	The PocketQube shall meet all dimensional and mass requirements specified in the PocketQube Standard Document. Parachute mass and volume is not included in the PocketQube mass and volume but is included in the overall CanSat mass requirement.	Verified during FRR via fit check using supplied 3D-printed fit gauge
PQ2	The PocketQube shall be 2p in length.	Verified during FRR
PQ3	The PocketQube shall include two microswitches/limit switches with levers that disconnect battery power when depressed by the deployment rails.	Verified during FRR
PQ4	The microswitches/limit switches shall be wired in series with the battery.	Verified during FRR
PQ5	The PocketQube shall include an umbilical interface that is easily accessed without removing any PocketQube structure to provide power for testing and/or battery charging. This is used when not stowed in the container.	Verified during FRR
PQ6	The PocketQube shall include a power switch or remove before flight switch that is accessible while integrating into the container prior to submission. The team shall remove the RBF or arm the PocketQube before turning in for rocket integration.	Verified during FRR
PQ7	The PocketQube shall include two side deployable solar panels of 2P length and at least 45 mm wide.	Verified during FRR
PQ8	The PocketQube shall include release mechanisms for the solar panels.	Verified during FRR
PQ9	The PocketQube shall include a boom with a magnetometer mounted on the end.	Verified during FRR

PQ10	The boom shall extend 220 mm from the nadir edge of the PocketQube when commanded by the ground station.	Verified during FRR
PQ11	The boom release mechanism shall be triggered by an uplink command.	Verified during FRR and observed via ground station during launch
PQ12	The boom shall be positioned so it shall be in the camera field of view when deployed.	Verified during FRR and reported by team during PFR using onboard recordings
PQ13	A recording video camera shall be included that points in the nadir direction.	Reported by team during PFR using onboard recordings
PQ14	The video camera shall have a minimum resolution of 640 x 480 pixels and be in color.	Reported by team during CDR
PQ15	The video of the ground shall be stabilized so the ground does not rotate during the flight.	Reported by team during PFR
PQ16	The video camera recording operation shall start when the PocketQube powers on.	Reported by team during PFR
PQ17	The video camera orientation shall be controlled by an uplink command.	Observed during launch
PQ18	The camera orientation command shall control the angle change in degrees.	Observed during launch
PQ19	The PocketQube shall support commands to trigger all deployment mechanisms.	Observed during launch and reported by team during PFR
PQ20	The PocketQube shall support simulation mode and accept air pressure data from the ground station and operate as flying.	Only verified if weather conditions preclude launch and testing must be done via simulated conditions. See weather delays
PQ21	The PocketQube shall be powered off while in the container.	Verified during FRR
PQ22	The PocketQube shall turn on when it exits the container.	Verified during FRR reported by team during PFR
PQ23	The PocketQube shall transmit	Observed on ground station during

	telemetry at 4 Hz once powered.	launch
PQ24	The PocketQube shall measure air pressure in pascals to 1 pascal resolution.	Reported by the team during PFR using onboard sensors
PQ25	The PocketQube shall measure internal temperature in C to 0.1 degree resolution.	Reported by the team during PFR using onboard sensors
PQ26	The PocketQube shall measure battery voltage in volts to 0.1 volt resolution.	Reported by the team during PFR using onboard sensors
PQ27	The PocketQube shall measure battery current in amperes to 0.001 amp resolution.	Reported by the team during PFR using onboard sensors
PQ28	The PocketQube shall measure the rotation rate in the X,Y,Z axis in degrees/sec to 0.1 degree/sec resolution.	Reported by the team during PFR using onboard sensors
PQ29	The PocketQube shall measure acceleration in the X,Y,Z axis in m/s to 0.001 m/s/s resolution.	Reported by the team during PFR using onboard sensors
PQ30	The PocketQube shall measure earth's magnetic field in the X,Y,Z axis in milli Gauss to 1 milli Gauss resolution.	Reported by the team during PFR using onboard sensors
PQ31	The PocketQube shall measure the solar panel voltages in volts to 0.1 volt resolution	Reported by the team during PFR using onboard sensors
PQ32	The PocketQube shall include a deployment detector for each solar panel.	Verified during FRR
PQ33	The PocketQube shall include a deployment detector for the boom indicating the boom is in the proper deployed position.	Verified during FRR
PQ34	The PocketQube shall track its position using a GNSS receiver.	Reported by the team during PFR using onboard sensors
PQ35	The PocketQube shall have an audio beacon that activates when the PocketQube turns on.	Verified during FRR

PQ36	The audio beacon shall not be software controlled.	Reported by team during CDR

Common Mechanism and Mechanical Requirements

Requirement Number	Requirement	Evaluation
M1	No pyrotechnical or chemical actuators are allowed.	Reported by team in PDR and verified during FRR
M2	Mechanisms that use heat (e.g., nichrome wire) shall not be exposed to the outside environment to reduce potential risk of setting the vegetation on fire.	Verified during FRR
M3	The PocketQube shall not be disassembled to reset any mechanisms.	Verified during FRR
M4	All mechanisms shall be capable of maintaining their configuration or states under all forces experienced during flight.	Observed during flight
M5	All electronics and mechanical components shall be hard mounted using proper mounts such as standoffs, screws, or high performance adhesives.	Reported by team in PDR and verified during FRR
M6	No deployable components shall use any container structure to hold in the stowed position.	Reported by team in PDR and verified during FRR
M7	The CanSat shall survive 15 Gs vibration	Verified during FRR via drop test
M8	The CanSat shall survive 30 G shock.	Verified during FRR via drop test
M9	Total mass shall be between 800 and 1000 grams which includes container, PocketQube, parachutes and any materials included with the CanSat.	Verified at CanSAT turn-in
M10	The Cansat Shall survive temperatures	Verified via environmental testing

	up to 60 C for up to 2 hours and not deform or become structurally compromised.	

Common Electrical Requirements

Requirement Number	Requirement	Evaluation
E1	Lithium polymer batteries are not allowed.	Reported by team at CDR and verified at FRR
E2	Battery source may be alkaline, Ni-Cad, Ni-MH or Lithium-ion. Lithium-ion cells must be manufactured with a metal package similar to 18650 or CR123 cells. Coin cells are allowed.	Verified during FRR
E3	Battery holders shall not use the spiral spring contacts.	Verified during FRR

Communications Requirements

Requirement Number	Requirement	Evaluation
X1	The ESPNOW communications protocol shall be used.	Verified during FRR
X2	Each team shall be assigned a channel to use and must set the channel before arriving at the launch site.	Verified during FRR
X3	The container and PocketQube radios shall use the ground station radio MAC	Verified during FRR

	address as the peer MAC.	
X4	The ground station radio shall receive only the team's telemetry.	Verified in PDR, CDR
X5	The ground station shall command only the team's PocketQube and container.	Verified in PDR, CDR
X6	No radio shall use the broadcast address.	Verified during FRR

Ground Station Requirements

Requirement Number	Requirement	Evaluation
GS1	The ground station shall include one or more displays to show the telemetry from the container and PocketQube.	Verified during FRR and observed during launch
GS2	For the container, the altitude, temperature, and battery current shall be plotted in real time.	Observed during launch
GS3	For the container, the non-plotted telemetry shall be displayed graphically or in easily readable text.	Observed during launch
GS4	For the PocketQube,the Altitude, temperature, battery voltage, battery current, PocketQube orientation in three axes based on sensor data,GPS position, and solar panel voltages shall be plotted.	Observed during launch
GS5	For the PocketQube, the non-plotted telemetry shall be displayed graphically or in easily readable text.	Observed during launch
GS6	The ground station shall send commands to the container and the	Evaluated during FRR and observed during launch

	PocketQube.	
GS7	The ground station shall support simulation mode.	Evaluated during FRR
GS8	Simulation mode shall read a file with an air pressure profile and send updated air pressure values at 1 Hz.	Only verified if weather conditions preclude launch and testing must be done via simulated conditions. See: weather delays
GS9	The ground station shall operate off of battery for at least two hours.	Reported by team during CDR
GS10	The ground station antenna shall be hand held or table top mounted.	Observed during launch
GS11	The ground system shall count the number of received packets. Note that this number is not equivalent to the transmitted packet counter, but it is the count of packets successfully received at the ground station for the duration of the flight.	Verified during FRR
GS12	Because the ground station must be viewed in bright sunlight, the displays shall be designed with that in mind, including using larger fonts (14 point minimum), bold plot traces and axes, and a dark text on light background theme.	Observed during launch
GS13	The ground station shall generate ACSII formatted csv files as specified in the Telemetry Requirements section. A separate file is required for container and PocketQube.	Verified during post-launch data review
GS14	Each team shall develop their own ground station.	Reported by team during PDR
GS15	The ground station shall be portable so the team can be positioned at the ground station operation site along the flight line. AC power will not be available at the ground station operation site.	Observed during launch
GS16	The ground station software shall be	Only verified if weather conditions

	able to command the payload to operate in simulation mode by sending two commands, SIMULATION ENABLE and SIMULATION ACTIVATE.	preclude launch and testing must be done via simulated conditions. See: weather delays
GS17	The ground station shall be able to activate all mechanisms on command.	Verified during FRR
GS18	The ground station shall not occupy more than one 6 foot long table.	Observed during launch

Flight Software Requirements

Requirement Number	Requirement	
F1	The flight software shall maintain a count of packets transmitted which shall increment with each packet transmission throughout the mission. The value shall be maintained through processor resets.	Verified during FRR
F2	The Cansat shall maintain mission time throughout the entire mission even in the event of processor resets or momentary power loss.	Reported by team during CDR; team should describe hardware/software solution being used to meet this objective
F3	The Cansat container shall have its time set by ground command to within one second UTC time prior to launch.	Observed during launch
F4	The flight software shall support simulated flight mode where the ground station sends air pressure values at a one second interval using a provided flight profile file.	Only verified if weather conditions preclude launch and testing must be done via simulated conditions. See weather delays
F5	In simulation mode, the flight software shall use the radio uplink pressure values in place of the pressure sensor for determining the payload altitude.	Only verified if weather conditions preclude launch and testing must be done via simulated conditions. See weather delays
F6	The flight software shall only enter	Only verified if weather conditions

	simulation mode after it receives the SIMULATION ENABLE and SIMULATION ACTIVATE commands.	preclude launch and testing must be done via simulated conditions. See weather delays
F7	The flight software shall include commands to activate all mechanisms. These commands shall be documented in the mission manual.	Verified during FRR
F8	Configuration states such as zero altitude calibration software state shall be maintained in the event of a processor reset during launch and mission.	Reported by team during CDR; team should describe hardware/software solution being used to meet this objective

Bonus Objective

Develop a science or engineering experiment to be added to the PocketQube. The science or engineering experiment can deploy on a boom or panel. It must meet all PocketQube requirements. The team designs the experiment that can be performed in the expected environment at the expected altitudes. Bonus point will be awarded if all

Requirement Number	Requirement	Evaluation
B1	The science experiment shall be functional in the expected environment reaching up to 670 m above ground level (AGL) and performed during descent.	Observed during flight and reported by the team during PFR
B2	The science experiment data or status information shall be transmitted in real time.	Observed during flight
B3	The science experiment shall not use any pyrotechnics, flames, radioactive materials, animals, insects, or hazardous chemicals. Personal protective apparel shall not be required when handling the experiment.	Reported by team during PDR

B4	Any materials, components, devices deployed in the experiment shall be tracked and recovered. Littering is not allowed.	Observed during flight
B5	Teams shall declare the number of items that will separate from the PocketQube in the performance of the experiment.	Reported by team during CDR and observed during flight

CanSat Competition Requirements

To ensure smooth running of the competition and maintain fairness amongst all participants, teams must comply with the deliverable and team requirements.

Requirement Number	Requirement	Evaluation
CS1	Each team shall consist of 3-10 students. The maximum number of students is dictated by the composition of grad and undergrad students as detailed in the Team Composition section.	Reported by team during application process
CS2	Teams shall deliver PDR charts, in PDF format, in accordance with the published deliverable schedule which fulfills the objectives discussed in Preliminary Design Review.	Verified by delivery timestamps
CS3	Teams shall participate in a 40-minute PDR during pre-coordinated time (Preliminary Design Review).	Evaluated by competition judges during PDR.
CS4	Teams shall deliver CDR charts, in PDF format, in accordance with the published deliverable schedule which fulfills the objectives discussed in Critical Design Review.	Verified by delivery timestamps
CS5	Teams shall participate in a 40-minute CDR during pre-coordinated time	Evaluated by competition judges during CDR.

	(Critical Design Review).	
CS6	Teams shall deliver PFR charts, in PDF format, via USB drive in accordance with the published deliverable schedule which fulfills the objectives discussed in Post Flight Review.	Verified by delivery timestamps
CS7	Teams shall participate in 20-minute PFR during pre-coordinated time (Post Flight Review)	Evaluated by competition judges during PFR
CS8	Presentation slides shall comply with the published Slide Format Guidelines	Evaluated by competition judges during PDR and CDR
CS9	Teams shall comply with Disqualification Guidelines	Evaluated by competition judges
CS10	Teams shall participate in 30-minute FFR during pre-coordinated time (Flight Readiness Review)	Evaluated by competition judges during FRR

Telemetry, Command, and Simulation Requirements

The container and PocketQube shall transmit telemetry and receive commands as specified below (Requirements CTR14 & PQ23). The telemetry and command formats between the ground station and the container and PocketQube are to be defined by the team. The format can be in binary or ASCII.

The telemetry submitted and reviewed after the mission shall be in ASCII CSV format (Requirement GS13).

The ASCII format of the telemetry packets to be submitted are described below. Each telemetry field is delimited by a comma, and each telemetry packet is terminated by a single carriage return character. No comma (‘,’) characters should be part of the data fields -- commas are delimiters only. The telemetry file shall be in CSV format and properly loadable into a spreadsheet program. Two telemetry files shall be provided, one for the container telemetry and a second for the PocketQube telemetry.

Science experiment data shall be appended to the end of the PocketQube telemetry data.

Container Telemetry

The container telemetry shall include what is described in the table .

Telemetry	Description
ID	Team Number + C for container. E.g. 1001C
MISSION_TIME	Time since power up, in seconds with 1 ms resolution.
PACKET_COUNT	Incrementing integer count of telemetry packets.
COMMAND_COUNT	Number of commands received in integer format.
MODE	Indicate ‘F’ for flight and ‘S’ for simulation mode.
ALTITUDE	Altitude in meters with 0.1 m resolution. (Requirement CTR19)
PRESSURE	Air Pressure in Pascals with 1 pascal resolution (Requirement CTR18)
TEMPERATURE	Temperature in C with 0.1 C resolution (Requirement CTR20)
BATTERY_VOLTAGE	Battery input voltage with 0.1V resolution (Requirement CTR21)
BATTERY_CURRENT	Battery input current with 1 ma resolution (Requirement CTR22)
MECH_STATE	Hex value with each bit used to indicate release Bit 0 - Deployment mechanism activated Bit 1 - PocketQube released
STATE	Flight operation state: **LAUNCH_PAD**- Waiting for command or launch **ASCENT**- detected rocket ascent **APOGEE**- detected peak altitude **PQ_RELEASE** - PocketQube was released
CMD_ECHO	Text of last command received and processed by the container. For example, CXON or SP101325. See the command section for details of command formats. Do not include commas characters.

Container Telemetry Format

The Cansat container telemetry CSV file format shall be as follows:

> **ID, MISSION_TIME, PACKET_COUNT, COMMAND_COUNT, MODE, ALTITUDE, PRESSURE, TEMPERATURE, BATT_V, BATT_I, MECH_STATE, STATE, CMD_ECHO**

Axes definitions for rate, acceleration, and magnetic sensor data:

> The following satellite industry standard definition of the Roll, Pitch, and Yaw axes shall be used for telemetry data and must be considered when mounting and aligning inertial measurement units.
>
> When perfectly aligned (zero pointing error on all axes), the positive Yaw axis (+Z) points towards the earth (nadir direction), the positive Pitch axis (+Y) points south, and the positive Roll axis (+X) points east.
>
> The camera pointing requirement is due north, which is the negative Pitch -Y direction.

PocketQube Telemetry

The PocketQube Telemetry shall include what is described in the table.

Telemetry	Description
ID	Team Number + P for PocketQube. E.g. 1234P
MISSION_TIME	Time since power up in seconds.
GNSS_TIME	GNSS Time in GNSS provided format.
PACKET_COUNT	Incrementing count of telemetry packets
COMMAND_COUNT	Number of commands received
CMD_ECHO	Text of last command received. Parameters not needed
MODE	Indicate ‘F’ for flight and ‘S’ for simulation mode.
ALTITUDE	Altitude of PocketQube in meters at 0.1m resolution derived from the air pressure sensor.
PRESSURE	Air Pressure in Pascals with 1 pascal resolution (Requirement PQ24)

TEMPERATURE	Temperature in C at 0.1C resolution (Requirement PQ25)
VOLTAGE	Battery voltage in V at 0.1V resolution (Requirement PQ26)
CURRENT	Battery current in mA at 1 mA resolution (Requirement PQ27)
ROT_RATE X,Y,Z	Rate gyro for each axis in degrees/sec at 0.1 degree/sec (Requirement PQ28)
ACCEL X,Y,Z	Acceleration in each axis in m/s/s at 0.001 m/s/s (Requirement PQ29)
MAG X,Y,Z	Calibrated magnetometer values in milliGauss at 1 mGauss (Requirement PQ30)
GNSS Position GNSS_LATITUDE GNSS_LONGITUDE GNSS_ALT	Latitude in degrees, longitude in degrees, altitude in meters at the resolution provided (Requirement PQ34)
GNSS_SATS	Number of satellite used in position solution (Requirement PQ34)
SOLAR_1	Solar panel 1 voltage in Volts at 0.01V resolution
SOLAR_2	Solar panel 2 voltage in Volts at 0.01V resolution
MECH_STATE	Hex value using bits to indicate each deployment mechanism status. (Requirement PQ33) Bit 0 = Solar panel 1 deployment state Bit 1 = Solar panel 2 deployment state Bit 2 = Boom deployment state Bit 3 = Camera recording on or off
IMAGE_STABILIZATION	Data from control loop showing stabilization operation. Number of fields is up to team control loop implementation.
SCIENCE_EXP	Science experiment data. Number of fields is up to the experiment.

PocketQubeTelemetry Format

The Cansat PocketQube telemetry CSV file format shall be:

ID, MODE,MISSION_TIME, PACKET_COUNT, COMMAND_COUNT, ALTITUDE, TEMPERATURE, PRESSURE, VOLTAGE, CURRENT, GNSS_TIME, GNSS_ALTITUDE, GNSS_LATITUDE, GNSS_LONGITUDE, GNSS_SATS, ROT_RATE_X, ROT_RATE_Y, ROT_RATE_Z, ACCEL_X, ACCEL_Y, ACCEL_Z,

MAG_X, MAG_Y, MAG_Z, SOLAR_1, SOLAR_2, MECH_STATE, CMD_ECHO, IMAGE_STABILIZATION,SCIENCE_EXP

Backup Radio

Since the competition is using the ESPNOW protocol for the first time, a back up radio is permitted to provide telemetry in case of issues with the ESP radio. The backup radio is not required.

If included, the back up radio shall be an XBEE radio operating in the 2.4 GHz or 900 MHz band. The NETID/PANID shall be set to the team number.

Telemetry Data Files

The received telemetry for the entire mission shall be saved on the ground station computer as comma separated value (.csv) files that will be examined by the competition judges in Excel. The CSV format should be the same as used by export from Excel.

Teams shall provide the CSV file to the judges immediately after the launch operations via USB drive.

The CSV files shall include a header specifying the name of each field/column of data in the file.

The telemetry data files shall be named as follows:

- **Flight_<TEAM_ID>C.csv** for the container telemetry**.**
- **Flight_<TEAM_ID>P.csv** for the PocketQube telemetry.

where the team_id is the four digit team id number. For example: Flight_1000C.csv is the required file name for imaginary team 1000 container telemetry.

The ground software shall produce the files, with the correct name, easily from the ground system user interface, and save them to the provided USB memory stick, which is to be given to judges before leaving the launch area.

On-board Telemetry Storage

It is suggested that teams make use of onboard data storage as backup in case of radio failure. Only the transmitted telemetry is examined and scored on flight day; however, the backup data can be used when preparing the Post Flight Review presentation.

Commands

The container and PocketQube shall support the following commands as a minimum. Additional commands are permitted. Teams can create their own command formats and protocols. (Requirement GS6, CTR28)

Command	Description
SIM	Set the device to simulation or flight mode.

SIMP	Send a simulated pressure data value.
CAL	Calibrate sensors. This command applies only to the container and is to be performed when the ground station judge asks for it to be performed.
ARM	Activate the launch detection for the container. This command is to be performed when the ground station judge asks for it to be performed.
MEC	Control a mechanism. This command will be used for testing, at FRR, and in case of an emergency, during flight.

The Cansat payload shall operate in two modes (Requirement GS7):

- FLIGHT mode where the Cansat operates as described in the Mission Overview section using actual sensor data, and
- SIMULATION mode where the Cansat receives simulated barometric pressure values from the Ground Station via SIMP commands and substitutes those values for the actual pressure sensor reading for calculation of altitude and for use by the flight software logic.

Simulation Mode does not affect sensor data other than the barometric pressure sensor. In Simulation Mode, the telemetered pressure sensor data should reflect the commanded simulation values, not the actual sensor readings.

The purpose of Simulation Mode is to provide a mechanism for testing the Cansat as well as for demonstrating the Cansat over the full mission profile in the event an actual launch cannot take place (due to weather or pandemic).

The team's Ground Station shall have the capability to read a barometric pressure data file that simulates the mission profile and transmit the values via commands to the Cansat at a rate of one per second (Requirement GS8). The pressure profile data file will be provided by the competition judges.

Simulation Mode Concept of Operation

For testing, pre-flight demonstration, and in a contingency where launch operations are not possible, a profile of barometric pressure data that represents the expected flight profile can be transmitted by the team's Ground System at a rate of one sample per second. Upon

receipt, the flight software will substitute the received value for the actual pressure sensor reading.

The Cansat telemetry operates the same in FLIGHT and SIMULATION mode. So, the values, other than the pressure and altitude (calculated from the pressure values) will be actual sensor readings (e.g., actual battery, temperature, and GPS).

Because accidental initiation of SIMULATION mode can endanger the flight mission, two commands, SIM ENABLE and SIM ACTIVATE, must be received by the flight software before Simulation mode is activated.

After activation, the flight software monitors the radio link for barometric pressure sensor commands (SIMP) from the Ground Station and uses the received values as if they were actual barometric pressure readings in the calculation of altitude, determination software state, and when to release the Cansat.

Environmental Tests

Four tests are to be conducted to test the construction quality and material performance. To verify test results, teams should provide: 1) Environmental Test Document based on the provided template file. 2) Videos of the tests performed as specified in the template document. If using a phone camera, orient the phone sideways for a wider video view. Only one submission will be accepted. Late submissions will not be scored.

1. Drop Test (Requirements M7 & M8) - This is to be performed on both the container with the PocketQube integrated and the PocketQube on its own.
   This test is designed to verify that the parachute and attachment point will survive the deployment. Component mounts and battery mount will also be tested. The drop test generates about 30 Gs of shock to the system.
   a. Drop Test Description - This test requires a 61 cm non-stretching cord. The test was developed with a 1/8 thick kevlar cord. One end is secured to an eyebolt attached to a fixed point, such as ceiling or rigid structure with enough clearance to accommodate the cord, Cansat, and free space so the Cansat does not hit the ground. The other end is tied to the parachute. A floor mat or pillow may be placed under the Cansat for the drop test. The structure must not flex during the drop test. This test cannot be performed by holding the cord. The cord must be secured to a solid structure. Holding any part of the test structure is not valid.
   b. Drop Test Procedure -

i. Power on Cansat.
ii. Verify telemetry is being received.
iii. Raise Cansat by the attached cord, so that the attachment points of the cord, on the eye bolt and the parachute, are at the same height.
iv. Release the Cansat.
v. Verify the Cansat did not lose power.
vi. Inspect for any damage, or detached parts.
vii. Verify telemetry is still being received.

2. Thermal Test (Requirement M10)- This test is to verify the Cansat can operate in a hot environment. When the Cansat is integrated into the rocket and sitting on the launch pad, the sun can heat up the payload section of the rocket to temperatures up to the mid to upper 40C. This test will determine if any materials warp, weaken, change characteristics, or fail to function at temperatures up to 60C.

a. Thermal Test Description - This test requires a method to heat the Cansat to 60C for a period of 2 hours. This will allow the components to rise to heat, and verify that they continue to function. One way to heat the Cansat is a thermal chamber; there are a few ways to build a thermal chamber.
   i. The simplest is to acquire an insulating cooler, one or more hair dryers and a thermometer, preferably a remote thermometer such as a thermocouple or thermistor. Place the Cansat, hair dryer(s) and thermometer in the cooler. Try to seal any openings. Make sure the hair dryer does not blow directly on the Cansat. The hair dryer will circulate and heat the air in the cooler.
   ii. Another method is to build a thermal chamber with foam insulation sheets. Select the sheets that have one side with a foil layer. Build a box and have one side be a lid. Seal all edges with duct tape. You can make a hole in the side to run cords and thermocouple through.
   iii. It is highly suggested the thermistor or thermocouple be attached to the Cansat. The purpose is to make sure the Cansat does reach the peak temperature. The air will reach the peak temperature long before the Cansat does.
b. Thermal Test Procedure -
   i. Place Cansat into a thermal chamber.
   ii. Turn on the Cansat.
   iii. Close and seal the thermal chamber.
   iv. Turn on the heat source.

v. Monitor the temperature and turn off the heat source when the internal temperature reaches 60C and turn on the heat source when the temperature drops to 55C.
vi. Maintain the test conditions for two hours.
vii. Turn off the heat source and perform visual inspection and any functional tests to verify the Cansat survived the thermal exposure and can operate as expected.
viii. With the Cansat still hot, test any mechanisms and structures to make sure the integrity has not been compromised. Take precautions to avoid injury.
ix. Verify epoxy joints and composite materials still maintain their strengths while at 60C.

3. Vibration Test (Requirements M7 & M8) - This test is designed to verify the mounting integrity of all components, mounting connections, structural integrity, and battery connections.

a. Vibration Test Description - The test uses a random orbital sander. The sander is a hand held power tool where the sanding head moves in a random pattern, not spinning. Random orbital sanders operate at fixed orbits per minute (opm) ranging from 12,000 to 14,000 opm. That translates to 200 to 233 Hz. This test takes advantage of the power up and power down phases of the sander. The sander does not instantly turn on at 14,000 opm, it takes at most a second to get up to speed. That transition time can hit some resonances of the Cansat. This test requires the sander to be cycled regularly over a one minute duration exposing the Cansat to vibrations from 0 Hz to 233 Hz. The amount of shaking generated by the sander is around 20 to 29 Gs. Do not use a disc sander that just spins.
    i. One setup for the test is to secure the sander, upside down, via a bench vise; the Cansat should be secured where the sand-paper is installed. The bench vise must be secured to a table. The vise cannot be allowed to move freely and the sander cannot be held by hand. The model number of the sander must be provided in the documentation.
b. Vibration Test Procedure -
    i. Power on the Cansat.
    ii. Verify accelerometer data is being collected.
    iii. Power up the sander.
    iv. Once the sander is up to full speed, wait 5 seconds.
    v. Power down the sander to a full stop.
    vi. Repeat steps iii to v four more times.

vii. Inspect the Cansat for damage and functionality.
viii. Verify accelerometer data is still being collected.
ix. Power down Cansat.

4. Vacuum Test (Requirements C3, CTR18 & PQ24) - This test is designed to verify deployment operation of the payload(s). A vacuum chamber can be simply constructed using a bucket or pail of 5 gallons or 18+ litres. A lid can be used or a ¼ inch or 6 mm thick sheet of polycarbonate can be placed on top of the bucket. Do not use acrylic as it can shatter. A vacuum cleaner or shop vacuum can be used to pull a vacuum.

i. Suspend the fully configured and powered Cansat in the vacuum chamber.
ii. Turn on the vacuum to start pulling a vacuum.
iii. Monitor the telemetry and stop the vacuum when the peak altitude has been reached.
iv. Let the air enter the vacuum chamber slowly and monitor the operation of the Cansat.
v. Collect and save telemetry
vi. Make the saved telemetry available for the judges to review.
vii. Show in video that all mechanisms are activated based on altitude changes, as applicable.

5. Impact Test - This test is designed to verify that the container will not structurally fail during flight and deployment so that it does not cause harm to the second CanSat in the rocket. The test shall be performed with the PocketQube integrated.

a. The container test requires a rubber sheet that is 0.5 inches or 12 mm thick. The durometer rating shall be 50A.
   i. Small Parts 33-005-500-012-012 General Purpose Rubber, 50A Durometer, Smooth Finish, No Backing, 1/2" Thickness, 12" Width, 12" Length, Black (purchased from Amazon)
b. Drop test procedure
   i. Turn on the container electronics.
   ii. Hold the container 20 cm above the rubber sheet with the door side facing down.
   iii. Let the container drop and free fall onto the rubber mat.
   iv. Check the container for damage.
   v. Verify the electronics are still functioning.
   vi. Repeat the drop with the opposite side facing the rubber sheet.
   vii. Verify the electronics are still functioning.
   viii. Verify the PocketQube was not damaged and still functional.

ix. Record the drops with video and show the container and PocketQube are still functioning.

Team Composition

Students currently enrolled in undergraduate degree programs, or students having graduated from such programs since the start of the current competition cycle, are counted as undergraduate students.

Students currently enrolled in post-graduate degree programs (MS, PhD), or students having entered such programs since the start of the current competition cycle, are counted as graduate students.

Team Size

Each team shall consist of between 3 and 10 students (undergraduate teams) from an accredited college or university (Requirement CS1). Teams may consist entirely of undergraduate students (undergraduate teams), entirely of graduate students (grad teams), or a combination of the two (mixed teams). Graduate teams shall consist of no more than 5 students. Mixed teams shall consist of no more than 7 undergraduate students and 3 graduate students.

Teams from the same school must develop their designs independently and **not** copy from other teams. Bulk purchasing of materials is allowed such as batteries and raw materials for construction. Sharing tools and services are allowed. Designs must originate from within the team.

No more than three teams from any one school will be accepted in the competition.

Faculty Advisor

Each team must have a faculty advisor. The role of the faculty advisor is to:

- Provide a point of contact for the team, both with the university and the competition.
- Aid teams with logistics such as arranging conference rooms, laboratory resources, etc.
- Providing general guidance throughout the competition.

The faculty advisor shall not:

- Make design decisions or direct recommendations.
- Participate in more than an oversight role during reviews.

Deliverable Items

Teams will be evaluated based on a series of deliverable items provided at various stages of the development. The deliverable items are selected to provide representative real-world milestones for tracking the Cansat development and ensuring team success.

Preliminary Design Review

The PDR is a “multi-disciplinary technical review to ensure that the system under review can proceed into detailed design, and can meet the stated performance requirements within cost (program budget), schedule (program schedule), risk, and other system constraints”. The Cansat PDR shall demonstrate (Requirement CS2) :

- An understanding of the Cansat mission requirements
- Allocation and derivation of system and subsystem requirements
- Definition of the Cansat concept of operations
- Overview of preliminary design that meets specified requirements
- Results of, or identification of, necessary trades to support preliminary design. While it is ideal to have completed trades prior to the preliminary design, it is not necessary.
- Results of, or identification of, necessary prototyping or testing efforts necessary to support or finalize the preliminary design.
- Preliminary budget
- Detailed development schedule

Preliminary design reviews shall be conducted via teleconference coordinated by the team lead(s) (Requirement CS3). The PDR presentations shall be less than 40 minutes in duration including time for questions. Presentation reviewers shall be permitted to ask questions during the presentation (i.e., questions are not held until the end of the presentation).

The PDR shall follow the presentation template posted on the Cansat Competition website (Requirement CS2) .

Critical Design Review

The CDR is “a multi-disciplined technical review to ensure that the system under review can proceed into system fabrication, demonstration, and test; and can meet the stated performance requirements within cost (program budget), schedule (program schedule), risk, and other system constraints” (Requirement CS4). The CDR shall demonstrate:

- All PDR level requirement TBDs and TBRs shall be resolved
- Refinement of the Cansat CONOP
- Results of detailed design and analysis for each subsystem
- Verification that detailed design meets system and subsystem level requirements
- Identification of subsystem and system level tests necessary for requirements verification
- Results of requirements verification tests completed to date
- Overview of mission operations
- Preliminary launch day sequence of events

- Revised budget
- Updated development schedule

Critical design reviews shall be conducted via teleconference coordinated by the team lead(s). The CDR presentations shall be less than 40 minutes in duration including time for questions. Presentation reviewers shall be permitted to ask questions during the presentation (i.e., questions are not held until the end of the presentation). Each section of the CDR shall be scored in accordance with the values listed in the outline. The CDR shall contribute to the total evaluation of the Cansat design according to the values listed in the section Evaluation and Scoring (Requirement CS5).

The CDR shall follow the presentation template specified in the "Cansat CDR Outline" document available on the Cansat Competition website. Extra material in the form of backup slides is permitted (Requirement CS4).

Post Flight Review

The PFR provides an assessment of flight operations and results of the demonstration flight. The PFR provides an assessment of successful and unsuccessful flight operations. The PFR shall provide:

- Overview of mission objectives and Cansat design
- Comparison of planned and actual CONOPS and SOE
- Raw and processed data from flight operations
- Failure analysis and assessment (for unsuccessful mission objectives)

Post Flight Reviews shall be conducted the day following the demonstration flight activities, unless flight operations are canceled due to weather. Presentations shall be limited to 20 minutes, including questions (Requirement CS8).

Each section of the PFR shall be scored in accordance with the values listed in the outline. The PFR shall contribute to the total evaluation of the Cansat design according to the values listed in the section Evaluation and Scoring (Requirement CS8).

Post Flight Review presentations shall be submitted by 9:30 AM to the judges (Requirement CS7). There will be two presentation rooms. Teams will be pre-assigned to the presentation room. Each team will be given a thumb drive. Teams are to install their PFR slides on the thumb drive and deliver the thumb drive to the designated location between 9 AM and 9:30 AM Sunday. Teams delivering after 9:30 AM will lose points. This is done to make sure all teams have the same amount of time to prepare for PFR.

Deliverable Submissions and Scheduling

All deliverable items, prior to launch weekend, shall be submitted to the competition email by

the dates listed in Table 1. All deliverable items shall be submitted in PDF format using the naming convention listed in Table 1 where # corresponds to the assigned team number for each team and v# is a unique revision number for the review package that can be used to track revisions. For example, a submission for imaginary team number 1000 of version 2 of the PDR package would be named **`Cansat2026_1000_pdr_v02.pdf`**. Note that adherence to the filename and format specification is scored during the competition. (Requirements CS2, CS4, & CS6)

If presentations need to be resubmitted, the version number must be incremented to ensure the most recent version is considered in scoring. With a large number of submissions and resubmissions, it is not possible to track correctly without using the version numbers.

Presentations will be scheduled after submission of the document. A calendar of available time slots will be sent to all teams. Each team is to send to the competition email a list of three time slots. The team will be notified of which time slot they are assigned.

Updated presentations will not be accepted after the deadline. The scoring is based on the quality of the presentation and understanding of the competition requirements.

Table 1: Deliverable item due dates

Material Due	Required Filename Format	Due Date
PDR	Cansat2027_XXXX_pdr_vYY.pdf	01/28/2027 23:59:59 UTC
CDR	Cansat2027_XXXX_cdr_vYY.pdf	04/01/2027 23:59:59 UTC
Environmental Test Document	Cansat2027_xxxx_env.pdf	5/21/2027 23:59:59 UTC
Flight Telemetry Data	Flight_XXXX_.csv for Science Payload.	06/12/27
PFR	Cansat2027_XXXX_pfr_vYY.pdf	06/13/2027 09:30:00 Local

XXXX is the team number. YY is the revision number. **Use this file format or your team will be removed from the competition. Files are to be in PDF format. No other formats will be accepted. Any submission after the deadline will be ignored.**

At the end of the competition, the PDR, CDR, and PFR packages may be placed on the website for reference in subsequent years.

Slide Format Guidelines

The following guidelines shall be used when developing the presentation material (Requirement CS8):

- Use the template made available. Failure to do so will result in loss of points.
- All slides shall have simple white backgrounds. This helps reduce the file sizes and makes the slides easier to read.
- All slides shall have page numbers in the footer. This is to allow for easier referencing of material during the reviews.
- All slides shall list the presenter's name in the footer. This provides all the reviewers with an identity as to who is presenting the material.
- No embedded files or movies shall be included in the presentations. Not all reviewers will be able to access or view movies during the reviews due to network security settings at the various organizations involved.
- Each line-item in the review outlines shall correspond to a dedicated slide. This may result in slides with single bullets on them, however, this makes it easier for the reviewers to follow the presentation.

A slide template is not provided for PFR. Teams are allowed to be creative in their slide design but must include all the elements provided which is lifted from the PFR scoresheet. Videos may be embedded. The PFR shall be self-contained and saved to the provided USB memory device (Requirement CS6) . Do not assume internet access will be available at the post flight review.

Disqualification Guidelines

Teams shall comply with the spirit and requirements of the competition (Requirement CS9) The following are grounds for removal from the competition.

1. Any team found to be copying a PDR/CDR document from a previous competition will be disqualified from the competition.
2. Any team not using the current mission guide will be disqualified.
3. Any team not meeting the basic competition requirements by CDR will be disqualified.
4. Any team submitting a PDR or CDR document that is scored at less than 70% will be disqualified.
5. Teams not demonstrating an understanding of the requirements of the competition will be disqualified.
6. Any team copying slides, content and designs from other teams will be disqualified.
7. Teams not submitting PDR and CDR documents in the required PDF format and file name convention as stated in the competition guide will be disqualified.
8. Teams not submitting PDR and CDR documents by the due dates will be disqualified.
9. Teams not scheduling presentation times by the specified date will be disqualified.
10. Teams not attending the PDR and CDR teleconference within 15 minutes of the set time will be disqualified. Find a phone that works. It can be a computer, landline or mobile phone.

11. Teams exceeding 10 members at any time during the competition.
12. Excessive arguing with judges or any staff.

Launch Weekend

Schedule

All times are referenced to eastern daylight time.

The competition starts Thursday and ends Sunday evening.

The preflight briefing will be held Thursday.

Flight readiness review and safety inspection will occur on Friday.

Saturday will be the launch day unless weather causes a postponement.

Sunday will be Post Flight Review presentations.

A detailed schedule will be provided at a later date.

Flight Readiness Review

On Friday, teams are required to have their Cansats inspected for flight worthiness. Each team will be assigned a one half hour time slot to present their Cansat. This means the Cansat must be completed, built and ready to launch at the flight readiness review. A ball of wires and boards does not constitute flight ready. To emphasize, when presented at the Flight Readiness Review (FRR), it must be in a state where it can be turned on and placed into a rocket for immediate launch and be fully operational. **Flight Readiness Review will be scored once. A team member must initial the score sheet to indicate agreement.**

FRR Sequence of Events

The test and verification sequence shall follow the FRR score sheet order. The ground station functionality and data display shall be verified. The mechanical structure shall be verified for safety. Crew assignments must be submitted at the flight readiness review in the Mission Operations Manual. The mission operations manual will be reviewed at the FRR. (Requirement CS10)

A separate station will be for fit check and drop test. A fit check will be performed first to verify the container slides in and out of the airframe freely. The Cansat must be in flight configuration and will be subjected to the drop test. If the test fails, the team must make repairs before being allowed to fly. The Cansat must pass the drop test in order to be launched. Two stickers will be added to the Cansat container indicating passing of the fit check and drop test.

Safety is the highest priority. Any Cansat deemed not flight worthy will not be flown. The team will lose all flight day points.

Team Member Launch Operations Crew Assignments

Crew assignments must be submitted at the flight readiness review. The mission control officer will be given an identification so the flight coordinator and launch control officer knows who the mission control officer is.

The mission operations manual will be reviewed at the flight readiness review.

Team Member Launch Operations Crew Assignments

In order to have a successful launch, teams need to coordinate among themselves and with the flight coordinator. Team members need to be assigned to specific tasks and develop a checklist for a successful flight. The following task assignments must be delegated:

Mission Control Officer - This is a single person who is responsible for managing the team at the time of the launch. This person must verify with the ground station crew everything is ready. This person will do the countdown starting at 5. The rocket will be launched after the count reaches 1.

Ground Station Crew - This is one or more persons who is responsible for monitoring the ground station for telemetry reception and issuing commands to the Cansat. Only the ground station crew should be at the ground station since there is limited space.

Recovery Crew - This is one or two persons only responsible for tracking the Cansat and going out into the field for recovery. The recovery crew is responsible for returning the Cansat to the judges at check-in with any required payload still inside. The Cansat cannot be disassembled before presenting the Cansat to the judges for review and scoring or no points will be awarded.

Cansat Crew - This is one or more persons responsible for preparing the Cansat ,integrating it into the rocket, and verifying its status.

Team members can take on multiple roles except for the Mission Control Officer. The Mission Control Officer should be coordinating all efforts and interacting with the flight coordinator as needed. It is highly recommended that a checklist be developed that steps the crews through the preparation, integration, and flight operations.

Crew assignments must be submitted at the flight readiness review.

Mission Operations Manual

Each team is required to assemble a mission operations manual. The mission operations manual includes five checklists/operations procedures to be created by the team. The checklists are for configuring the ground station, preparing the Cansat, and integrating the

Cansat into the rocket. The launch preparation procedures, launch procedure, and removal procedure are to be provided. Additional steps can be added by the team. The document is available for download and modification. Each section of the mission operations manual must start on its own page. Pages should be numbered and a table of contents is to be included.

The team must have the mission operations manual assembled into a three ring binder. The mission control officer must use the manual during launch. The mission control operator shall go to the microphone at the launch site and announce their team and go through their launch procedure which will include the countdown to the launch.

Launch Schedule

All times are local time.

The launch will start at 1pm local time. All Cansats are to be submitted by noon. The time period is from 8am to 12pm local time and is available for launch preparations and check in. This is the time to prepare antennas, ground stations, and final Cansat tests and preparations.

Cansats must be submitted at the check-in by noon. Only one or two team members must be in line with the completed Cansat by noon. Teams who submit Cansats after the noon deadline will lose **100 points** from launch day points. Cansats must be in flight ready condition in order to be in line. Teams cannot be in line while working on the Cansat. The Cansat must be in the stowed configuration and off when submitted. Teams will be kicked out of the check-in line if they are seen working on their Cansat or their Cansat is not fully assembled in the stowed configuration.

The launch will start at 1pm and will be done in groups of five. Each team will be assigned a round which will be scheduled in one half hour increments. 15 minutes before the launch round, the teams assigned to the round shall have their CanSat turned on by a judge and installed in a rocket. Two CanSats will be placed in one rocket. Cansats shall not be disassembled at this stage. The Cansat must be flight ready and the only thing that can be done to the Cansat is to turn on the Cansat with the power switch. Any team who does not launch in their scheduled launch round will lose an additional 50 points. Any team that requires them to unload their Cansat from the rocket after the rocket is mounted on the launch pad will lose an additional **50 points**. Any team that requires disassembly of their Cansat will lose an additional **100 points**.

Each team will be given a thumb drive to upload their ground station data after they have completed their flight operations. The thumb drive must be submitted to the judging table before leaving the field.

There will be three judges per flight. A ground station judge will review the ground station and step through the ground station judging procedure. Once completed, the team will review the score and initial. A second judge will be the flight observation judge who tracks the Cansat and verifies the Cansat separates from the rocket, the payload releases some time afterwards and the descent control device activates and works properly. A team member must be with the judge and together verify the events and initial the score sheet. A third judge will be in a nearby tent to score the csv telemetry file. The team must save the data and then

present the csv file to the judge. The team member will show the judge the required data to be scored. The team member shall initial the score sheet when judging is completed.

Competition Operations and Sequence of Events

Details of flight day operations shall be provided at the Pre-Flight Brief. An overview of the flight day operations include the following activities:

1. Arrive at launch site
2. Prepare Cansat for turn in. Make it flight ready and perform any tests.
3. Turn in Cansat at the check-in table by noon. It will be weighed and fit checked and stored in the stowed configuration in the off state until rocket preparation time.
4. Upon the team round, the team will have their Cansat turned on by a judge and installed in the rocket.
5. Verify the Cansat is communicating with the ground station.
6. Take the rocket with the ground station to the assigned launch pad. A staff member will install the rocket on the launch pad.
7. When it is time to launch, a judge will come by the ground station to monitor the ground station operation.
8. The team mission control officer will go to the launch control table and execute the launch procedures with the flight coordinator providing oversight.
9. Ground station crew will perform all required flight operations such as turning on telemetry and calibrating the sensors.
10. After all Cansats have launched for the current half hour round, team recovery personnel can head out to recover.
11. Ground station crew must clear out of the ground station area to allow the next round ground stations to set up.
12. A ground station crew member must turn in the thumb drive with any ground station data to a designated judge and review the data for onsite scoring. The team member must initial the score sheet indicating agreement with the results. The thumb drive shall contain only the telemetry received from the ground station. Videos are to be presented at PFR.
13. Recovery crew must return the Cansat to check-in to complete any final judging requirements.

Second Flight Rules

Second flights are rare but do occasionally happen. The following conditions will qualify for a second flight.

1. Rocket engine failure, engine failure causes parachute not to deploy.
2. Rocket parachute failure. The parachute never deployed from the rocket.
3. CanSats tangled causing failure to deploy PocketQubes.

A team whose Cansat is destroyed due to a rocket failure will receive all launch day points only if they meet the following requirements:

1. passed the drop test and the fit check.
2. received all points on the FRR test.
3. Transmitted telemetry during ascent and descent.
4. Telemetry data showed all the required states until rocket failure.
    a. Ascent
    b. apogee
    c. descent
    d. Operation during descent.

If telemetry does not show the minimal required states, only points up to failure will be awarded.

Weather Delays

If the Saturday launch gets rained out, Cansat demonstrations shall be performed Saturday using the launch schedule. The demonstration will require the drop test followed by the simulation demonstration where simulated pressure data is transmitted to the Cansat (Requirements PQ20, GS8, GS16, F4, F5, & F6, M7).

If Sunday is clear, the launches will be performed and the PFR will be canceled.

If Saturday and Sunday are rained out, the demonstration will be performed Saturday and the PFR will be on Sunday.

Appendix A Field Safety Rules

1. Consumption of alcohol is not allowed.
2. Smoking is only allowed at designated areas. If anyone is caught smoking where it is not allowed, the landowner can throw you off the field.
3. Do not catch rockets or Cansats out of the air.
4. Stay behind the designated range line unless the range safety officer (RSO) or launch control officer (LCO) or flight coordinator has given permission for you to put your rocket on your assigned pad.
5. Pay attention at all times. Every launch is potentially hazardous.
6. If a “heads up” launch is announced, you must be standing and facing the launch pad.
7. Do not retrieve a rocket from the range unless the LCO has given you permission to do so.
8. Everyone must be alert when a “heads up!” is called and be ready to move.
9. Do not litter. Do not throw trash on the ground anywhere on the field. We have been invited to use the land owner's field and should treat it with respect. Any team caught throwing trash on the ground anywhere will be disqualified from the competition and the school will be notified of the disqualification. The landowner can order the team to leave the property and enforce the order.
10. No student or team drones or flying vehicles are allowed.

Appendix B Presentation Recommendations

The following recommendations for presentation content and layout are being provided based on past experiences of the judges. These recommendations are not required to be followed but make it easier for the judges to review the material presented.

1. Use a consistent table format throughout the various subsystems when presenting requirements, component trades, and changes since previous reviews. Using a standard table format makes it easier for the judges to find the information in the table quickly since all tables are formatted the same.
2. During the CDR, the Changes Since PDR slides should use a table that contains a discussion of what the state of the design was at PDR, what it is at CDR, and what the rationale of the change was. Details of the change can be discussed in subsequent slides so an in-depth discussion is not always necessary.
3. Include the class year (freshman, sophomore, etc.) and major of each team member for reference. This doesn't play into the scoring of the team, however, it is often nice for the reviewer to know the status of the team members.
4. Be sure to follow the PDR and CDR outlines very carefully. Provide at least one chart for each scored item in the outline; this makes it easier for the judges to follow the presentation and confirm the required information is provided. In the presentation, be sure to address the questions and topics listed in the "description" column of the presentation outline -- those are the key points the judges are looking for.
5. Be clear which optional requirements, if any, are to be included in the design.
6. Be detailed in test descriptions. Identify specific tests, what is going to be done, and the pass/fail criteria.
7. Include slide/page #s on all presentation pages.

Appendix C Payload Deployment Description

The Cansat will be mounted on top of the payload section with the required shoulder inserted into the airframe of the payload section along with any other part of the Cansat.

The rocket motor has a delay grain that starts burning when the rocket motor is ignited. The delay grain burns for 10 seconds on a J425 Aerotech motor and then ignites the black powder charge that forces the rocket to separate into two parts and deploy the rocket's parachute. At this time, the Cansat should be pushed out or fall out of the payload section and start its descent. The rocket will descend on its own parachute.

Appendix D Acronyms

A	Analysis
CDR	Critical Design Review
CONOP	Concept of Operations
D	Demonstration
DCS	Descent Control System
FRR	Flight Readiness Review
GCS	Ground Control Station
HW	Hardware
HWR	Hardware Review
I	Inspection
LCO	Launch Control Officer
PDR	Preliminary Design Review
PFB	Pre Flight Briefing
PFR	Post Flight Review
RPM	Revolutions Per Minute
RSO	Range Safety Officer
SOE	Sequence of Events
T	Test
TBD	To Be Determined
TBR	To Be Resolved
VM	Verification method

Appendix E Definitions

Analysis	Verification method that utilizes evaluation of data generated by accepted analytical techniques or simulations under defined conditions to show the item will meet the specified requirements.
CDR	A multi-disciplined technical review to ensure that the system under review can proceed into system fabrication, demonstration, and test; and can meet the stated performance requirements within cost (program budget), schedule (program schedule), risk, and other system constraints.
CONOP	Describes what the system will do and the way the system works from the operator's perspective. The CONOP is a high level description that should include a top-level block diagram.
Demonstration	Verification method that utilizes a qualitative exhibition of functional performance, usually accomplished with no or minimal instrumentation.
Inspection	Verification method that utilizes an examination of the item against applicable documentation to confirm compliance with requirements.
Need Date	Latest date a component or element (software, etc.) must be received or completed in order to not impact the end completion date.
PDR	A multi-disciplined technical review to ensure that the system under review can proceed into detailed design, and can meet the stated performance requirements within cost (program budget), schedule (program schedule), risk, and other system constraints.
Shall	Verb used to indicate a requirement is binding. All shall statements require verification.
Should	Verb used to define a goal or non-mandatory provision.
Test	Verification method utilizing operation of all or part of the item under controlled conditions, either real or simulated, to determine that the quantitative design or performance requirements have been met.
To Be Determined	An item or parameter that has not been specified at the time of document release.
To Be Resolved	An item or parameter that is preliminary or uncertain at the time of document release and for which a final value is to be specified at a later time.
Validation	Confirms that the system, as built (or as it will be built), satisfies the user's needs. Confirmation you built the right thing.
Verification	Confirms that the system, its elements, its interfaces, and incremental work products satisfy their requirements. Confirmation you built the

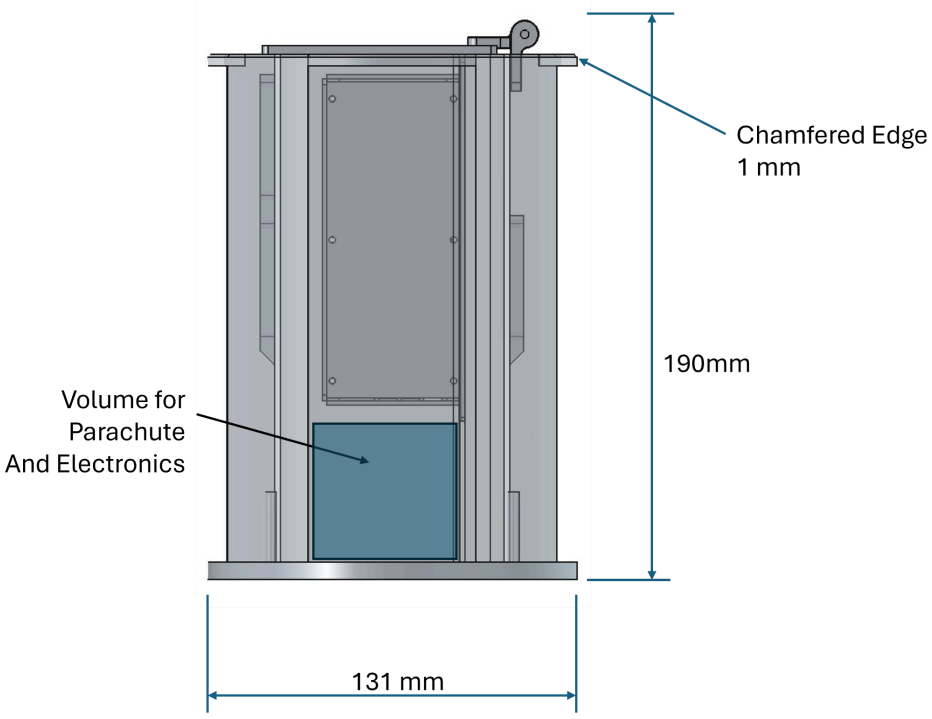
	system right.
Will	Verb used to reference a binding or hard requirement elsewhere in the document text.

Appendix F – Payload Section

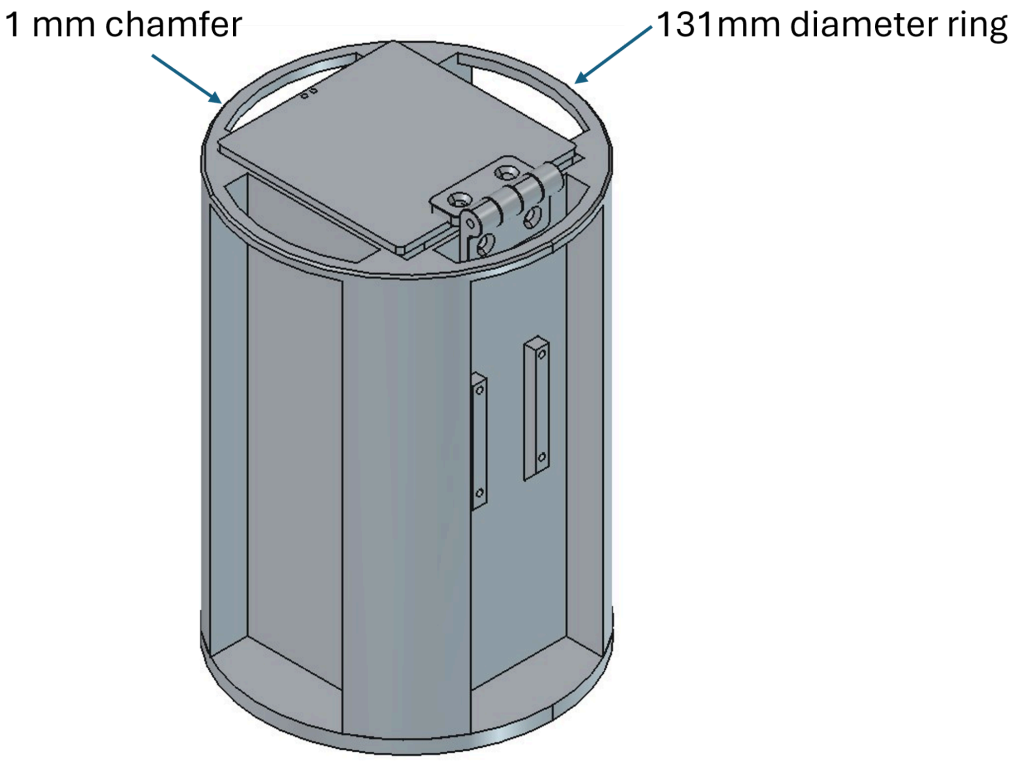
The Cansat shall consist of two parts, the payload and the container. The container is defined here. (Requirements CTR1, CTR2, CTR4, & CTR9)

Spec	Description
1	Volume diameter of 131 mm.
2	Volume length of 190 mm.
3	Top outside edges chamfered with 1 mm radius.
4	Top side to include a 131 mm ring or cylinder to keep the container from tilting in the rocket payload section.
5	PocketQube released from the top side.
6	Bottom and top side to have a 6 mm (¼") thick plywood disk. (Requirement CTR29) The top plywood disk does not need to be a solid disk. It can have an opening to allow the PocketQube to be released. The top disk shall have a continuous ring along the perimeter.
7	Plywood disk to be secured to the container structure with at least four bolts.
8	For the bottom disk where the parachute will be attached, two 5 mm holes opposite of each other by 20 mm centered at the disk center. (Requirement CTR30)
9	Position the PocketQube to get the center of gravity closest to the center of the cylinder volume.

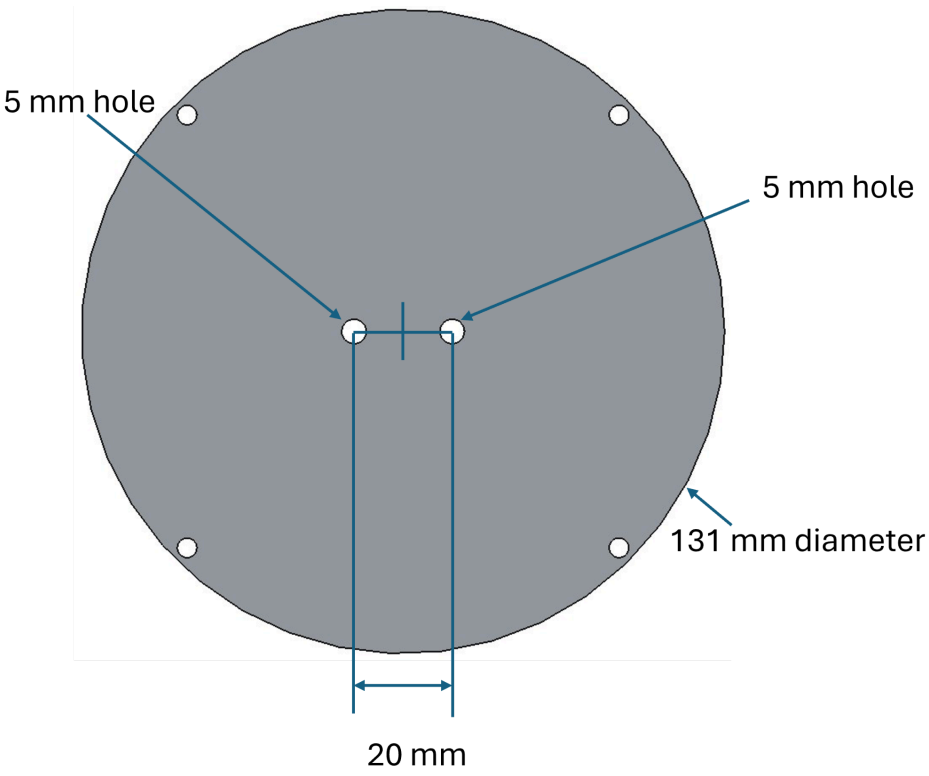
Images below are for reference only. They show the required dimensions.



Container Dimensions



Container Top Dimensions



Container Bottom Disk Dimensions



Two CanSats in Rocket Payload Section

Link to the rocket kit:

https://www.locprecision.com/product/minie-mag/

Link to payload section that is added to the kit:

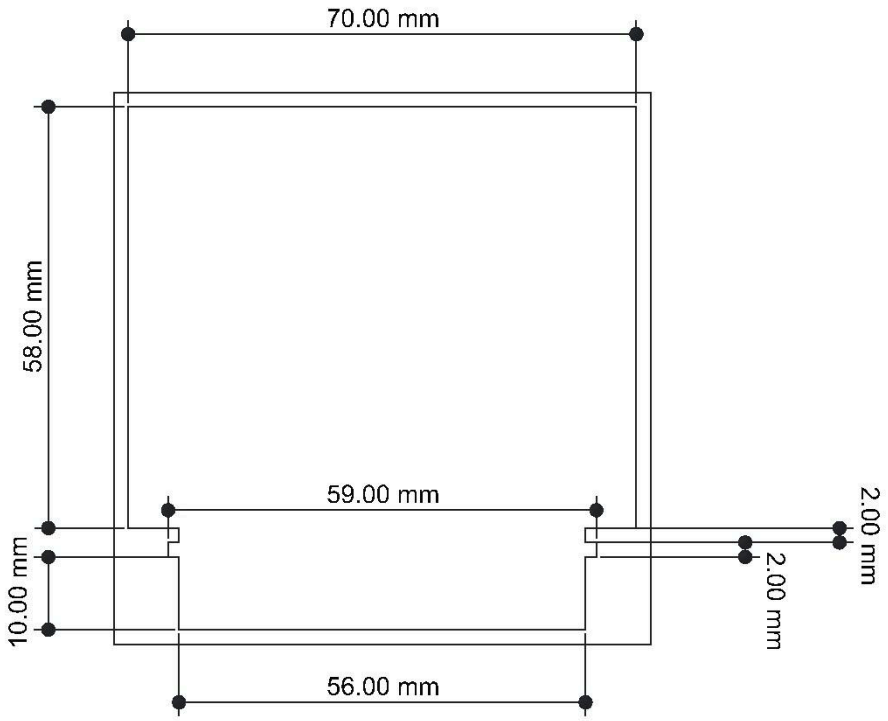
https://locprecision.com/collections/rocket-components/products/payload-bays?variant=39778637512895

Select 5.38” x 24” size

There are no other modifications to the rocket.

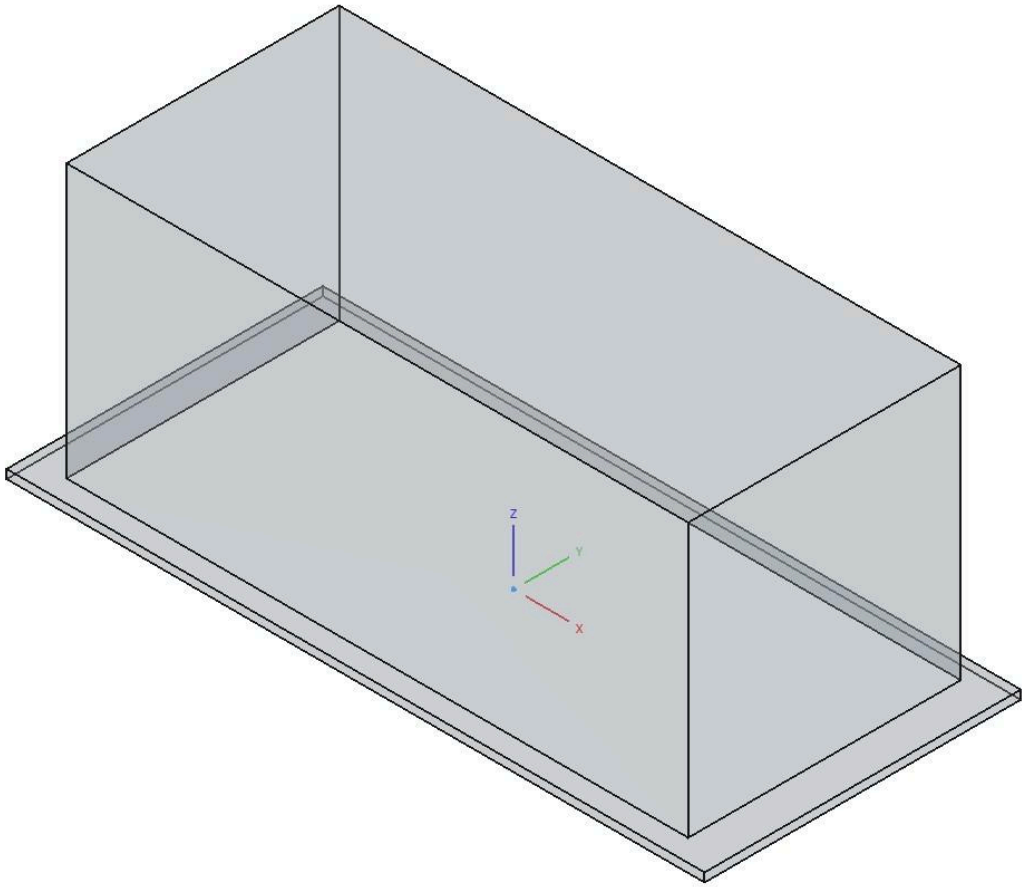
The rocket motor used will be an Aerotech J425.

Appendix G - PocketQube Fit Check Dimensions



Cross Dimensions

PocketQube with stowed deployables must fit within the dimensions shown.
The length of the PocketQube with deployables cannot extend beyond the 128 mm length.
(Requirements PQ1 & PQ2)



Coordinate System

Appendix H - Competition Calendar

Date	Action
8/12/2026	Competition Guide is posted.
9/1/2026	Application is open to teams.
11/1/2026	Applications are closed.
11/6/2026	Team selection for countries with more than 20 teams and payments open
12/15/2026	Application payments closed.
1/11 - 1/28/2027	PDR Time slot selection.
1/28/2027	PDR documents due by 23:59:59 UTC.
2/1/27-2/19/27	PDR presentations.
2/26//2027	Score sheets reviewed and sent to teams
2/26-3/5	Teams review and identify errors
3/10/2027	The top 50 teams are selected and invitation letters are sent out.
3/19 - 4/1/2027	CDR Time slot selection.
4/1/2027	CDR documents due by 23:59:59 UTC.
4/5/26-4/23/27	CDR presentations.
5/7/2027	CDR results sent to teams.
5/21/2027	Environmental Test Documents Due.
6/10/2027	Preflight Briefing
6/11/2027	Flight Readiness Review
6/12/2027	Launch Date
6/13/2027	Post Flight Review, Awards